

LESSON 7-6 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Distance Between Points on a Grid**

Maryland Standard 6.NS.C.8 · Learning target – I can find the distance between two points that share a coordinate.

REMEMBER

- **Same x-coordinate:** the points are on a vertical line. Use the y-coordinates.
- **Same y-coordinate:** the points are on a horizontal line. Use the x-coordinates.
- Opposite signs: **add** the absolute values. Same signs: **subtract** them.

WORKED EXAMPLE

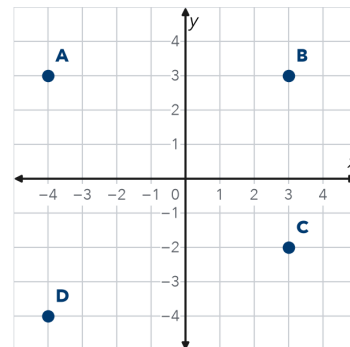
Find the distance from (3, -4) to (3, 5).

Step 1 Same x, so use the y-coordinates -4 and 5.**Step 2** Opposite signs, so add: $4 + 5 = 9$.**Answer 9 units****WORD BANK****distance** – how far apart; never negative**x-coordinate** – the first number; left or right**y-coordinate** – the second number; up or down**absolute value** – how far a number is from 0**PRACTICE***show your work · label units*

1. Use the grid. Find each distance.

Step 1 Find the coordinate that stays the same.**Step 2** Look at the other coordinate. Different signs? **Add**.
Same sign? **Subtract**.

- a. A to B _____ units
b. B to C _____ units
c. A to D _____ units



2. Find the distance between each pair of points.

- a. (5, 6) and (5, -2) ____ b. (-7, 1) and (-2, 1) ____
c. (-3, -4) and (6, -4) ____ d. (0, -8) and (0, 3) ____
e. (-9, -2) and (-9, -7) ____ f. (2.5, 3) and (-4, 3) ____
g. (-1, -6) and (-1, 4) ____ h. (-8, 5) and (-3.5, 5) ____

3. **City Blocks.** Kim walks **straight** from $(-6, -3)$ to $(-6, 5)$. Each unit is 1 block. How far does she walk?

Step 1 The x-coordinates match, so use the y-coordinates -3 and 5 .

Step 2 The signs are different, so **add** the absolute values.

Answer _____ blocks

4. **Missing Point.** Point P is at $(2, -3)$. Point Q is 6 units to the **left** of P. Name point Q.

Step 1 Moving left changes only the x-coordinate.

Step 2 Subtract 6 from the x-coordinate. Keep the y-coordinate.

Answer _____

5. **Find the Error.** Eli says the distance from $(1, -6)$ to $(1, -2)$ is 8 because $6 + 2 = 8$. Explain his mistake and find the correct distance.

Frame Eli is wrong because both y-coordinates are ____, so he should _____. The distance is _____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME With someone at home, count the blocks between two places on one street.

LESSON 7-6 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Comparing Distances on the Plane**

Maryland Standard 6.NS.C.8 • Learning target – I can decide whether to add or subtract to find a distance, then compare two distances.

REMEMBER

- Find **each** distance first. Then compare.
- Different signs: **add**. Same sign: **subtract**.
- A distance is never negative.

WORKED EXAMPLE

Which is longer: $(-2, 4)$ to $(5, 4)$, or $(1, -3)$ to $(1, 5)$?

Step 1 First: same y , use x : $2 + 5 = 7$ units.

Step 2 Second: same x , use y : $3 + 5 = 8$ units. $8 > 7$.

Answer the second one

WORD BANK

compare – decide which is greater

closer – a shorter distance

farther – a longer distance

PRACTICE

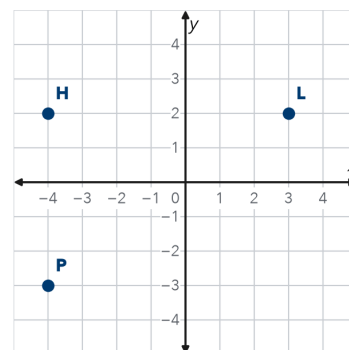
show your work • label units

- 1. Neighborhood.** Each unit is 1 block. Which is closer to Home: the Library or the Park?

Step 1 Find H to L. Then find H to P.

Step 2 Compare the two numbers. Subtract to find how many more.

- a. H to L _____ blocks
b. H to P _____ blocks
c. Which is closer? By how many blocks? _____



- 2.** Find each distance.

- a. $(-8, 1)$ to $(-3, 1)$ _____
b. $(4, -6)$ to $(4, 7)$ _____
c. $(-6, -7)$ to $(-6, -1)$ _____
d. $(-2.5, 3)$ to $(4.5, 3)$ _____

- 3. Runners.** Ana runs from $(-5, -1)$ to $(6, -1)$. Jay runs from $(2, 4)$ to $(2, -6)$. Who runs **farther**? By how many units?

Step 1 Find Ana's distance.

Step 2 Find Jay's distance.

Step 3 Subtract the smaller distance from the larger one.

Answer _____

- 4. Reasoning.** For $(-3, 7)$ to $(-3, -2)$, do you add or subtract? Explain, then find the distance.

Frame I ___ because the y-coordinates 7 and -2 have ___ signs. The distance is ___ units.

TEST PRACTICE

choose the best answer

- 5.** Segment JK runs from $(-4, 2)$ to $(6, 2)$. Segment MN runs from $(1, -3)$ to $(1, 5)$. Which is longer?

Step 1 Find the length of JK.

Step 2 Find the length of MN. Then compare.

- A** JK, by 2 units **B** MN, by 2 units **C** neither **D** JK, by 8 units
- 6.** Point A is at $(-2, -5)$. Point B is at $(-2, 3)$. What is the distance from A to B?
- A** 2 units **B** 3 units **C** 8 units **D** -8 units

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

AT HOME With someone at home, pick two places on one street. Which is closer to your home? By how many blocks?